

Characterising Network Traffic

6CS029 Advanced Networking

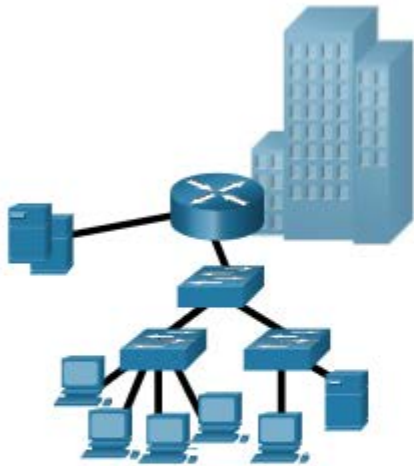


The Need to Scale the Network

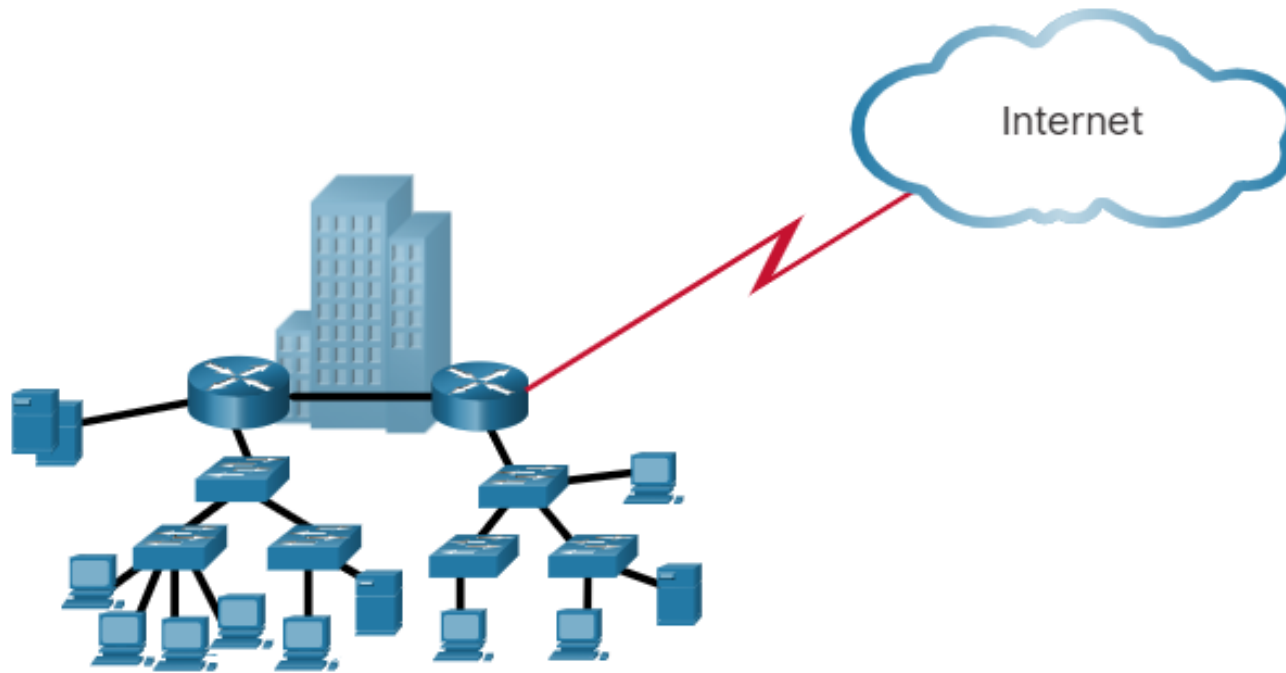
- Organizations increasingly rely on their network infrastructure to provide mission-critical services.
- Evolving organizations require networks that can scale and support:
 - Converged network traffic
 - Critical applications
 - Diverse business needs
 - Centralized administrative control
- Network designs include small networks that use a single LAN switch, up to very large networks with thousands of connections.



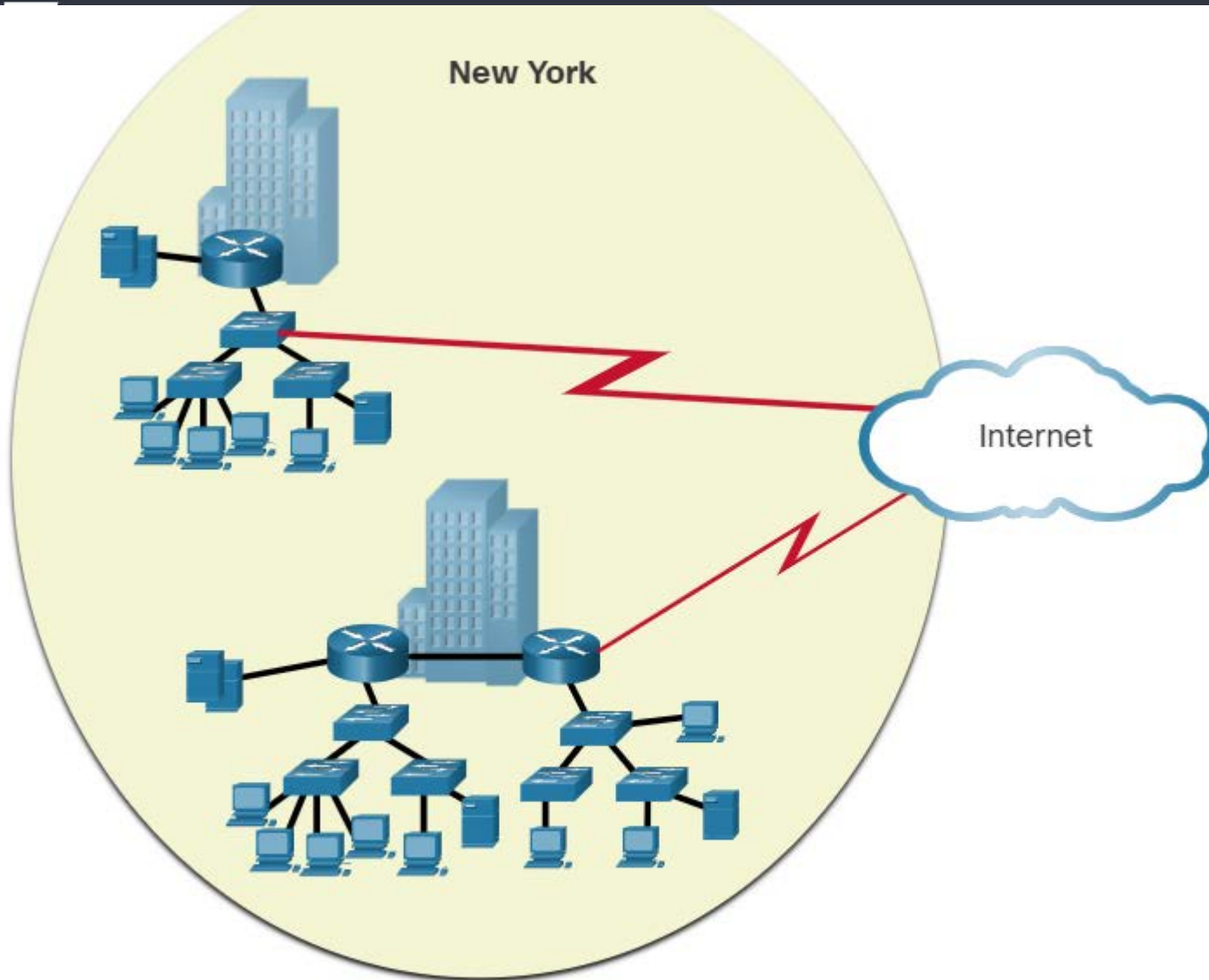
The Need to Scale the Network



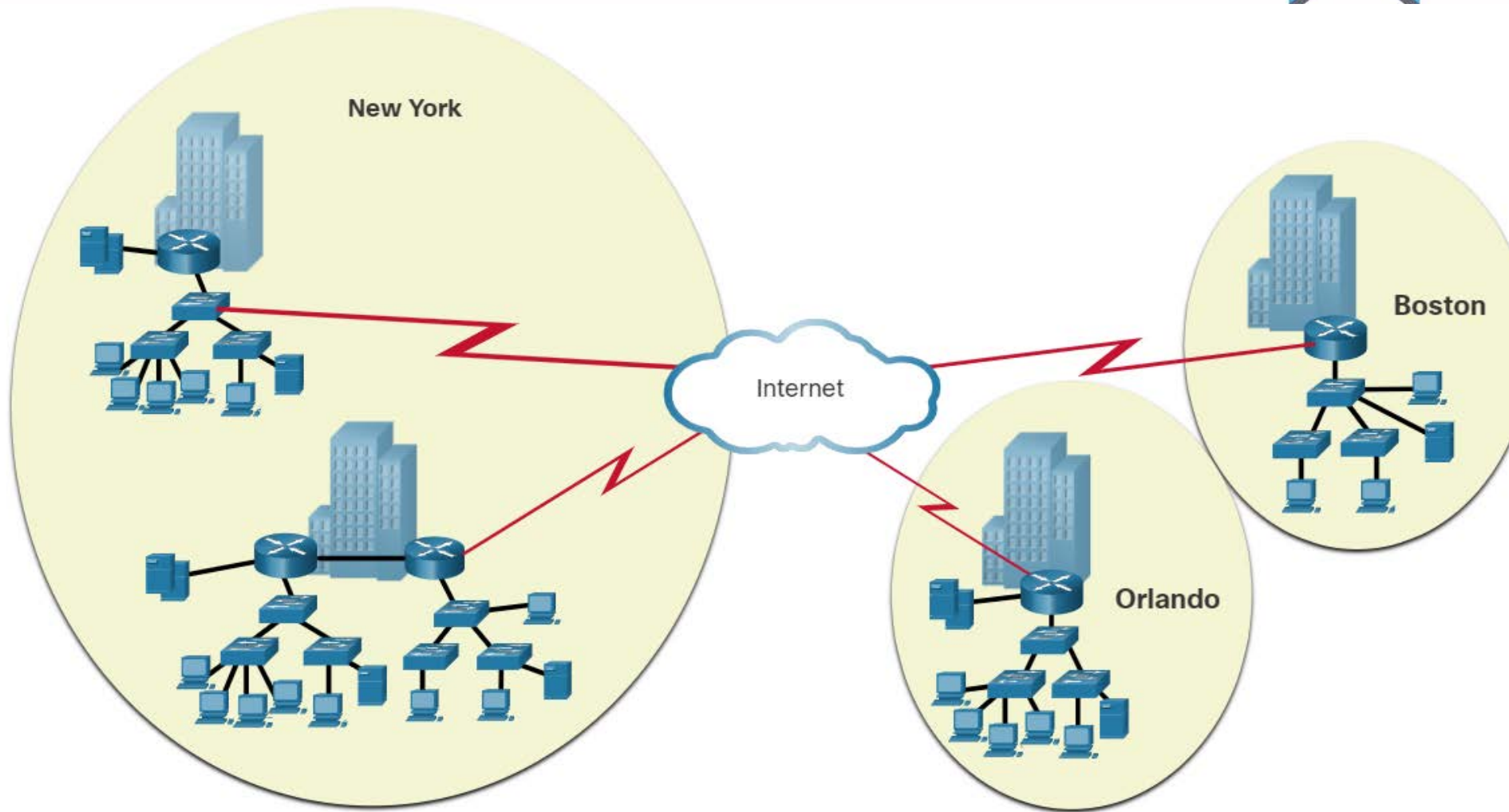
A small, single location company.



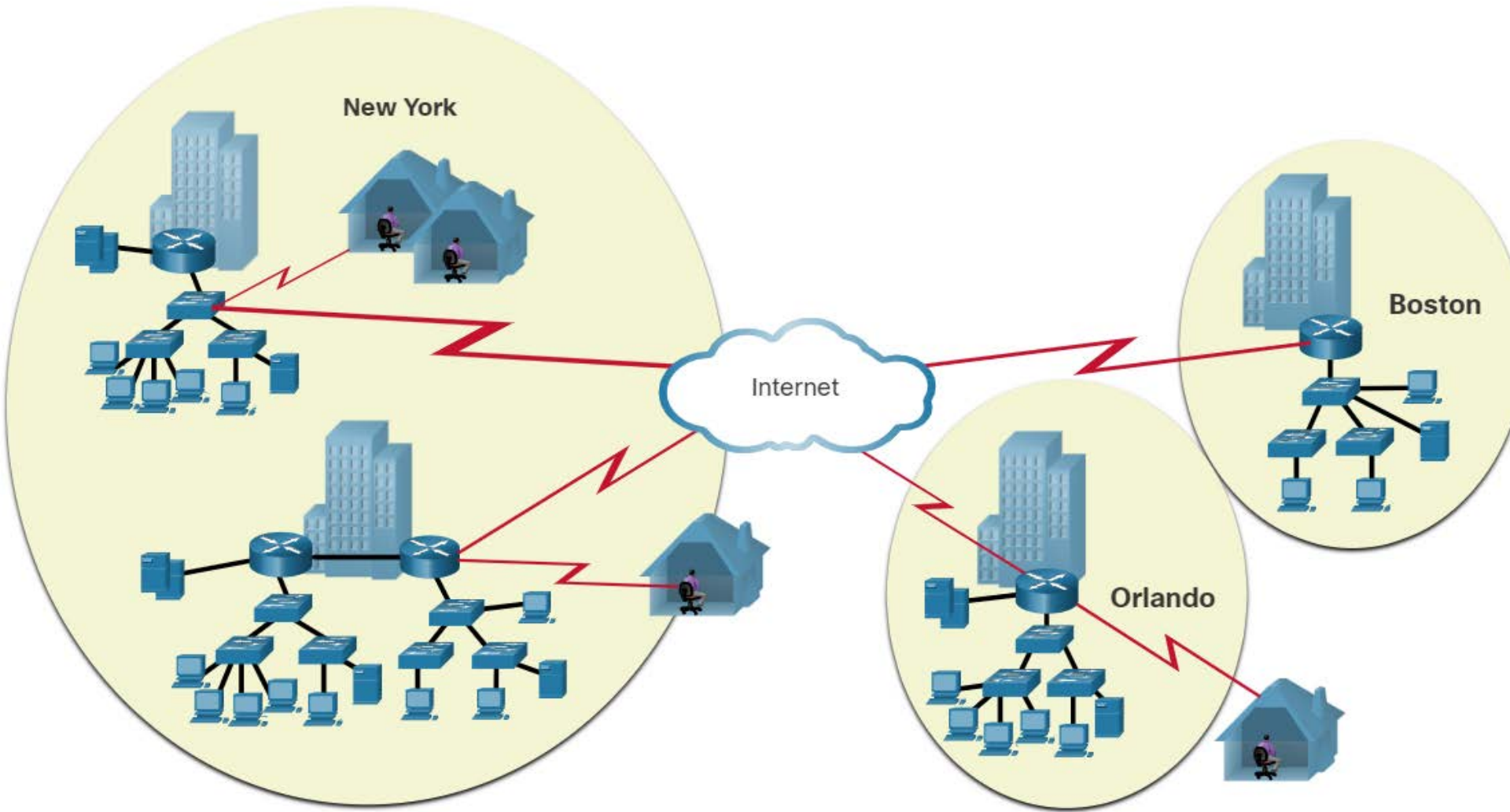
The company increases its number of employees.



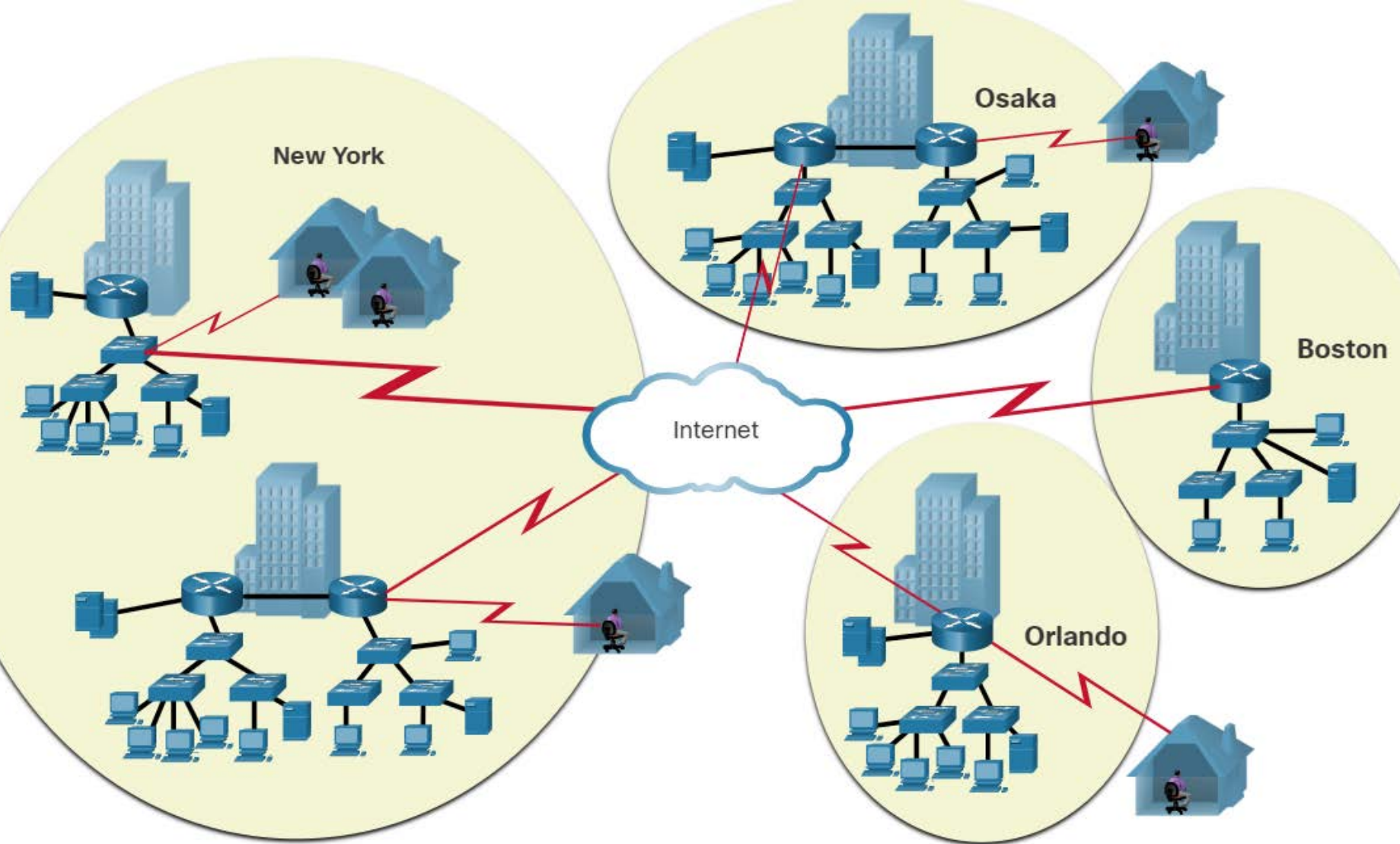
The company grows to multiple locations in the same city.



The enterprise grows to multiple cities.



The enterprise hires teleworkers.



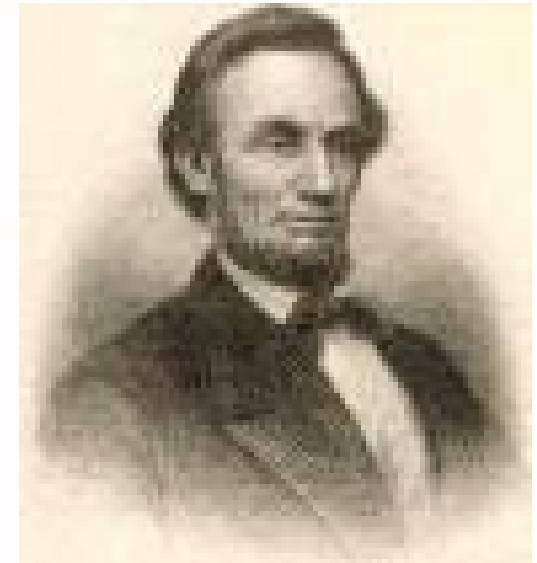
The enterprise expands to other countries.



Existing Network

What's the Starting Point?

*"If we could first know where we are
and whither we are tending, we
could better judge what to do and
how to do it."*



Abraham Lincoln



Existing Network

How do we evaluate the health of the existing network?

- Performance
- Availability
- Bandwidth utilization
- Accuracy
- Efficiency
- Response time
- Status of major routers, switches, firewalls, servers



Characterising Network Traffic

- Understand how network users interact with network resources and services
- Identify all user groups
- Assess the impact of adding new user groups



Characterising Network Traffic

What are the most important traffic factors to be taken into consideration?

- Traffic flow
- Location of traffic sources and data stores
- Traffic load
- Traffic behaviour
- Quality of Service (QoS) requirements



User Communities

User Community Name	Size of Community (Number of Users)	Location(s) of Community	Application(s) Used by Community



Data Stores

Data Store	Location	Applications	Used by User Communiti es



Traffic Flow

	Destination 1 MB/sec	Destination 2 MB/sec	Destination 3 MB/sec	Destination n MB/sec
Source 1				
Source 2				
Source 3				
Source n				

Traffic Flow

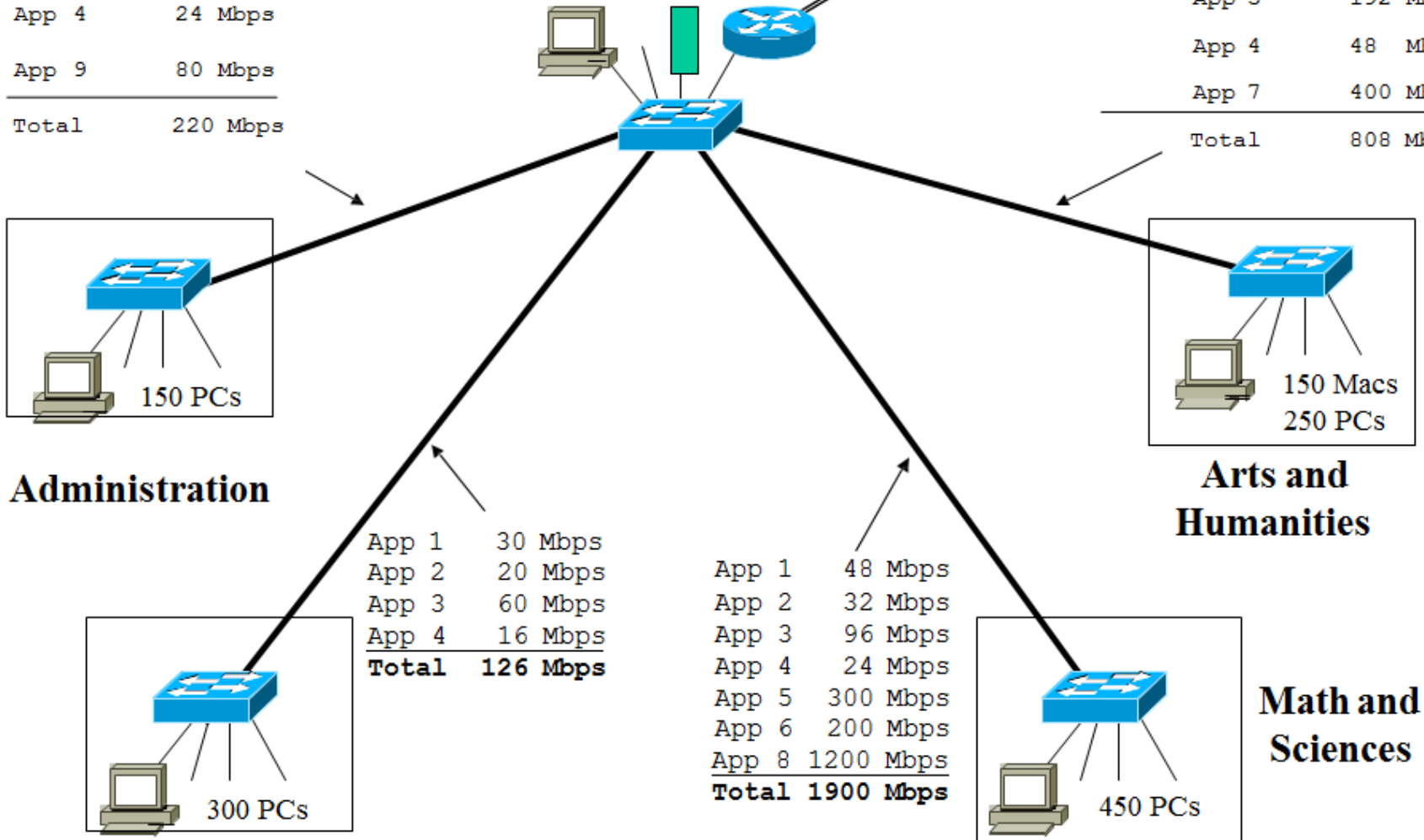
Library and Computing Center

100 Mbps MetroEthernet
to Internet

App 2	20 Mbps
App 3	96 Mbps
App 4	24 Mbps
App 9	80 Mbps
Total	220 Mbps

App 1	108 Mbps
App 2	60 Mbps
App 3	192 Mbps
App 4	48 Mbps
App 7	400 Mbps
Total	808 Mbps

Computing centre and ServerFarm



Administration

Arts and Humanities

Business and Social Sciences

Math and Sciences



Types of Traffic Flow

- Terminal/host
- Client/server
- Thin client
- Peer-to-peer
- Server/server
- Distributed computing



QoS Requirements

What makes traffic flow in voice over IP networks challenging to characterise and plan for?

- The flow associated with transmitting the audio voice is separate from the flows associated with call setup and teardown.
 - The flow for transmitting the digital voice is essentially peer-to-peer.
 - Call setup and teardown is a client/server flow
- QoS requirements for multimedia traffic

Name of Application	Type of Traffic Flow	Protocols Used by Application	User Communities That Use the Application	Data Stores (Servers, Hosts, etc)	Approximate Bandwidth Requirements	QoS Requirements



Traffic Load

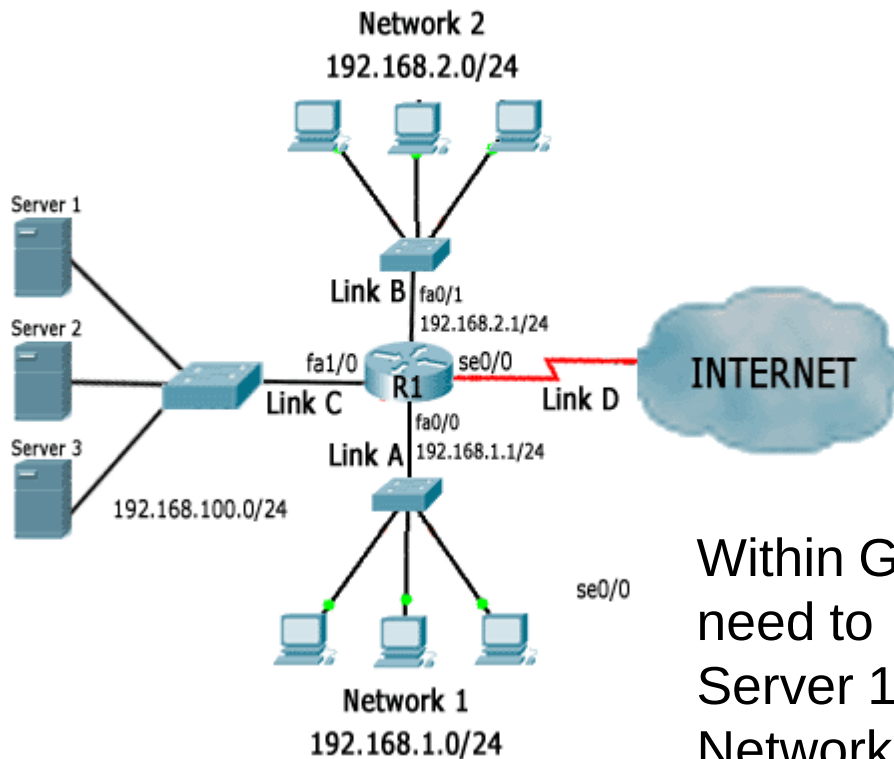
- Required capacity
 - The number of stations
 - The average time that a station is idle between sending frames
 - The time required to transmit a message once medium access is gained
- Traffic behaviour
 - Broadcasts
 - Multicasts



Typical Bottlenecks?

- Shared services (centralised servers etc)
- Multi-user applications and databases
- Low-speed NICs
- Shared LAN segments
- Low-bandwidth WAN links
- Core routing and switching components
- Firewalls (particularly public-facing)
- Inappropriate compression usage

Green Inc. Network Topology

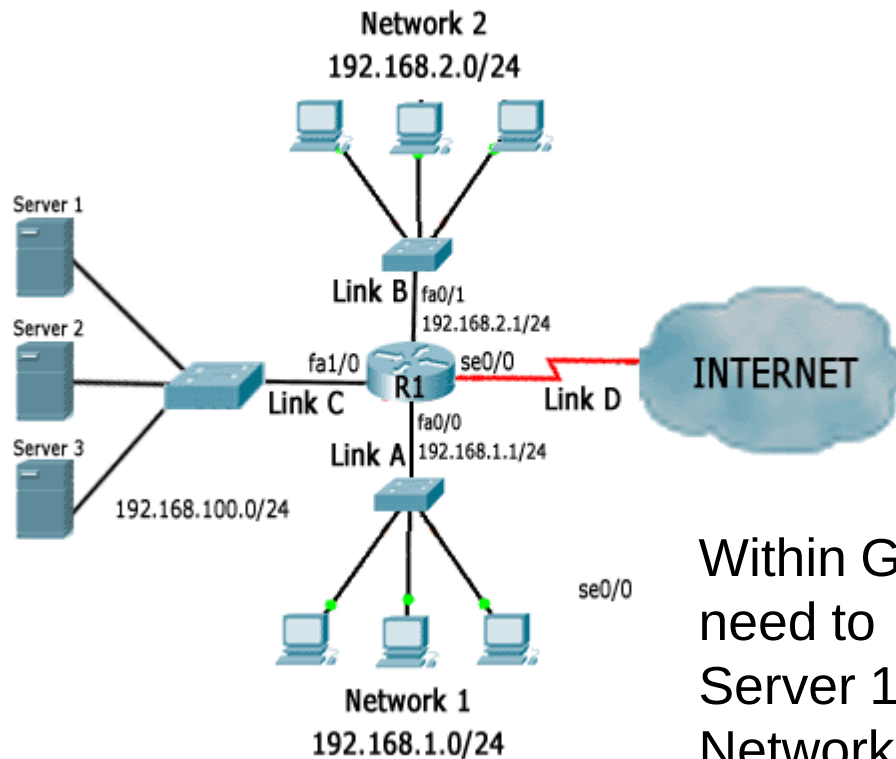


Green Inc. is expanding and just got another floor on the building they have their main office. The new floor will also need to be connected to Green's network.

Within Green Inc. Network, network 1 users need to frequently access a database stored in Server 1.
Network 2 users also frequently access a database stored in Server 2.

Task1: Identify performance, scalability and redundancy problems with this network design.

Green Inc. Network Topology



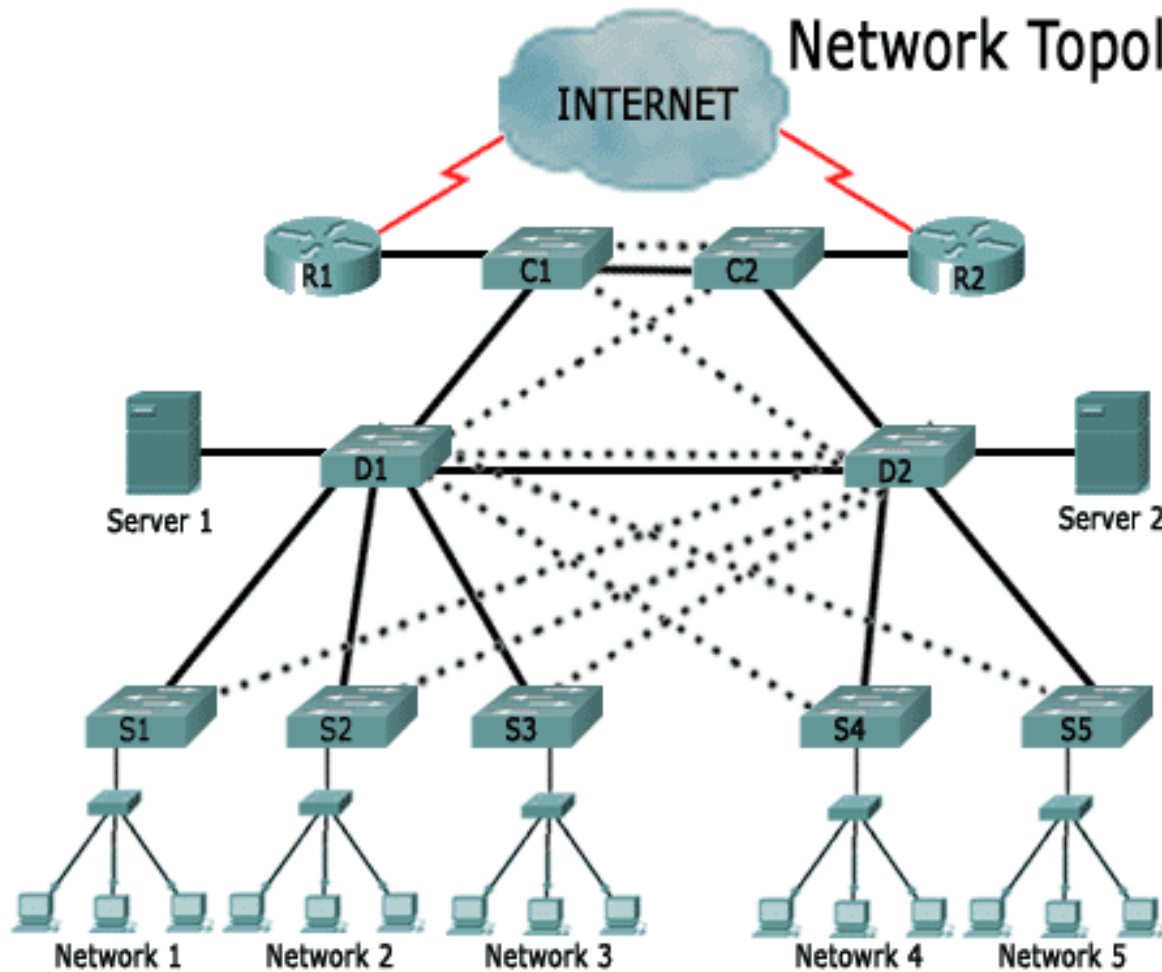
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Within Green Inc. Network, network 1 users need to frequently access a database stored in Server 1.
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Task2: Propose a new network design to solve the problems identified



Green Inc. Network Topology





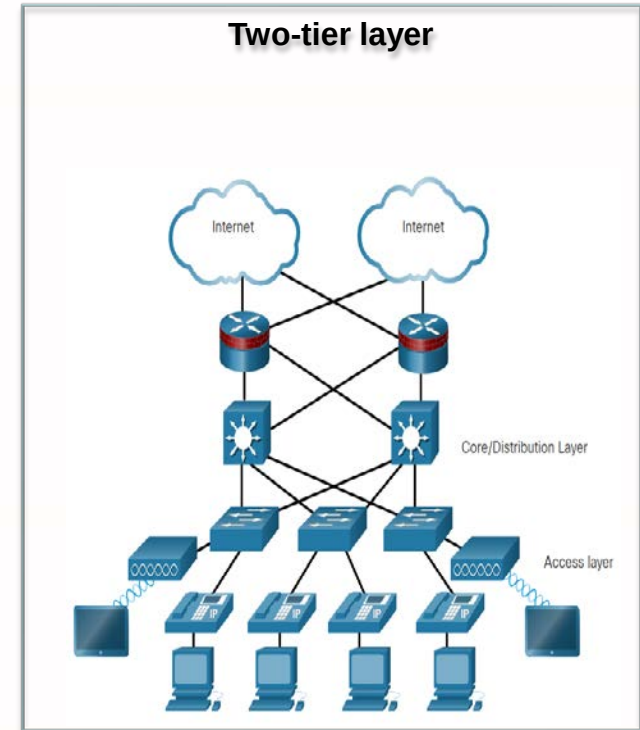
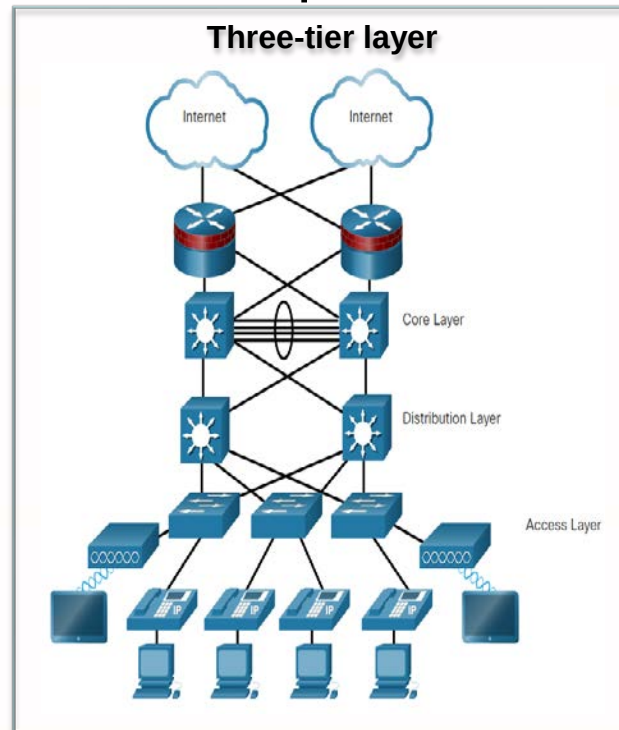
Borderless Switched Networks

- A network architecture that can connect anyone, anywhere, anytime, on any device; securely, reliably, and seamlessly.
- It provides the framework to unify wired and wireless access, built on a hierarchical infrastructure of hardware that is scalable and resilient.
- Borderless switched networks are hierarchical, modular, resilient, and flexible.



Hierarchy in the Borderless Switched Network

Hierarchical networks use a tiered design of access, distribution, and core layers with each layer performing a well-defined role in the campus network.





Access, Distribution, and Core Layer Functions

Access Layer

- The access layer provide network access to the user.
- Access layer switches connect to distribution layer switches.

Distribution Layer

- The distribution layer implements routing, quality of service, and security.
- It aggregates large-scale wiring closet networks and limits Layer 2 broadcast domains.
- Distribution layer switches connect to access layer and core layer switches.

Core Layer

- The core layer is the network backbone and connects several layers of the network.
- The core layer provides fault isolation and high-speed backbone connectivity.



Advantages of Hierarchical Network Design

- Scalability
 - Easier to add to the network
- Manageability
 - Easier to identify location of problems
- Broadcast traffic segmentation
 - Traffic confined to smaller broadcast domains
 - Less traffic over expensive links

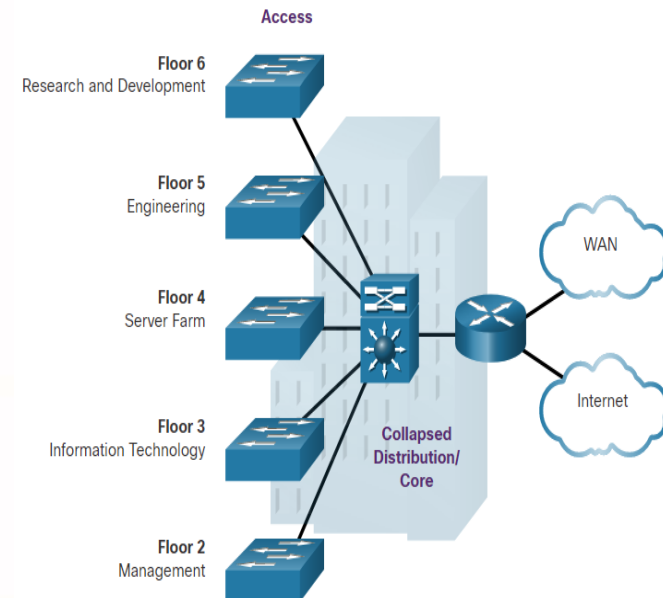
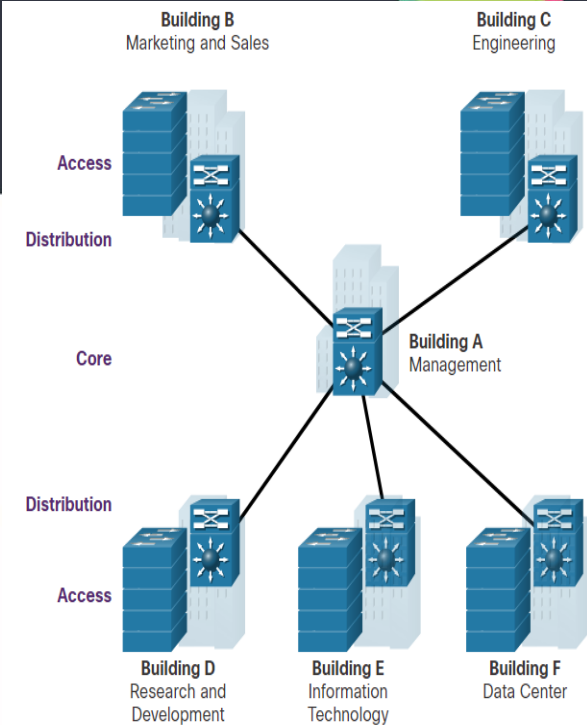
Three-Tier and Two-Tier Examples

Three-tier Campus Network

- Used by organizations requiring access, distribution, and core layers.
- The recommendation is to build an extended-star physical network topology from a centralized building location to all other buildings on the same campus.

Two-tier Campus Network

- Used when separate distribution and core layers is not required.
- Useful for smaller campus locations, or in campus sites consisting of a single building.
- Also known as the *collapsed core* network design.





Design for Scalability

Scalability is the term for a network that can grow without losing availability and reliability.

Network designers must develop strategies to enable the network to be available and to scale effectively and easily.

This is accomplished using:

- Redundancy
- Multiple Links
- Scalable Routing protocol
- Wireless Connectivity



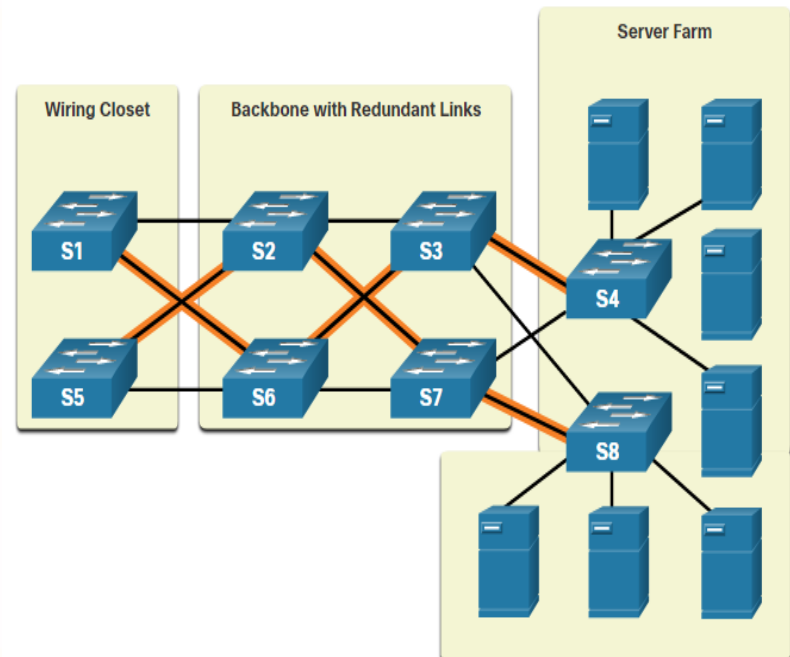
Plan for Redundancy

Redundancy can prevent disruption of network services by minimizing the possibility of a single point of failure by:

- Installing duplicate equipment
- Providing failover services for critical devices

Redundant paths offer alternate physical paths for data to traverse the network supporting high availability.

- However, redundant paths in an Ethernet network may cause logical Layer 2 loops.
- Therefore, Spanning Tree Protocol (STP) is required.

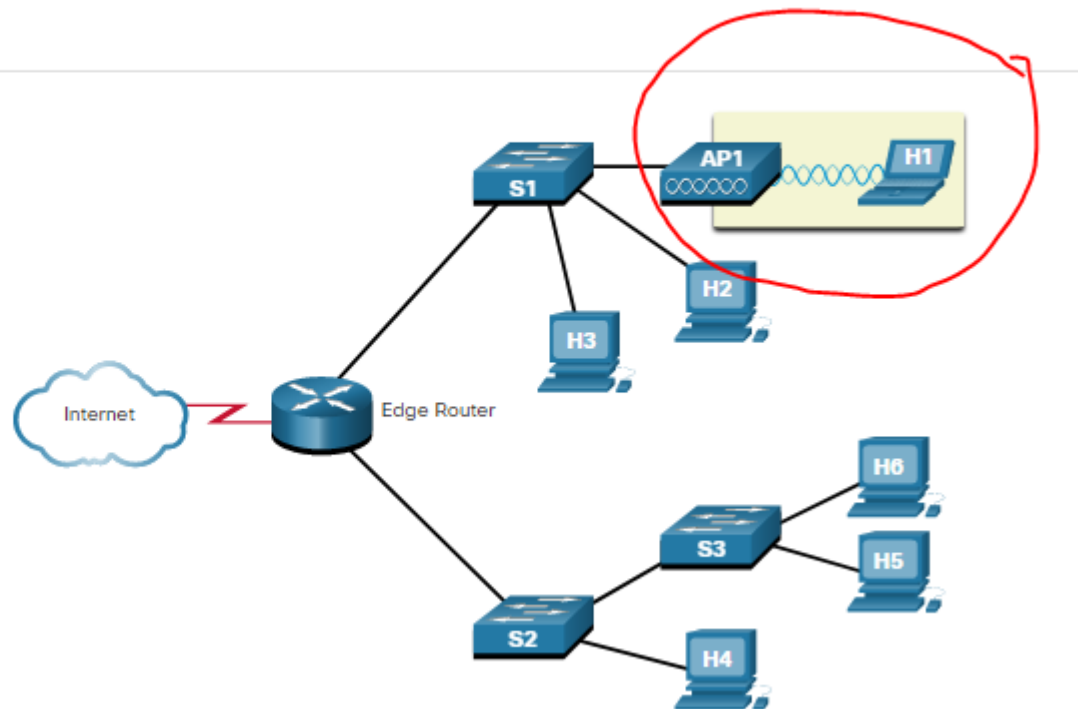




Reduce Failure Domain Size

- Edge Router
- AP1
- S1
- S2
- S3

AP1





Edge Router

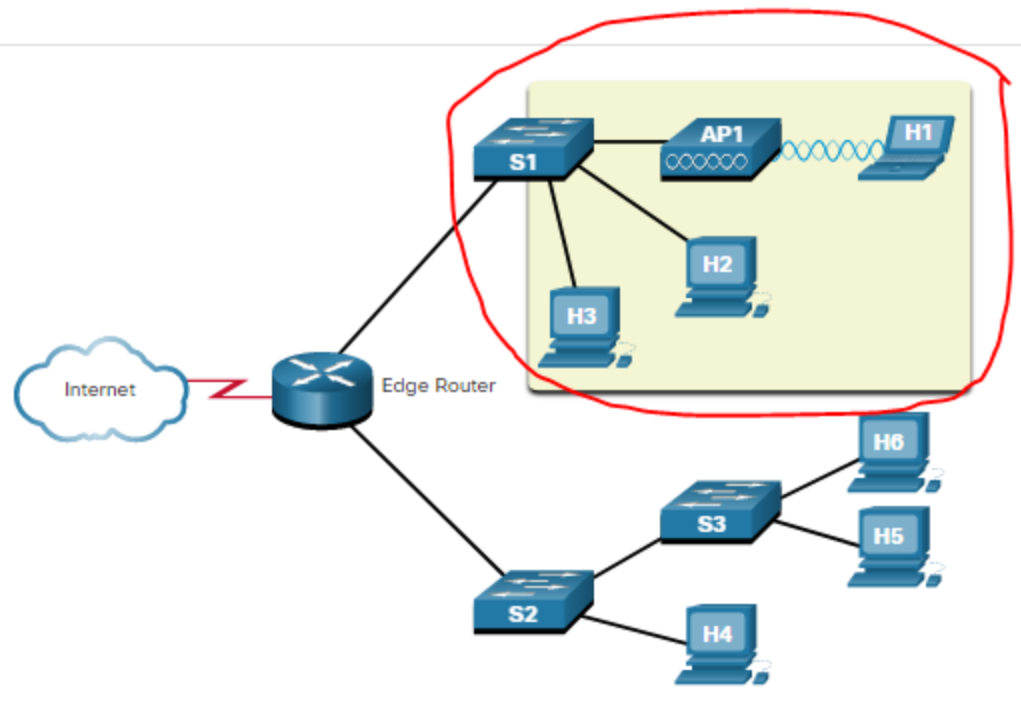
AP1

S1

S2

S3

S1





Edge Router

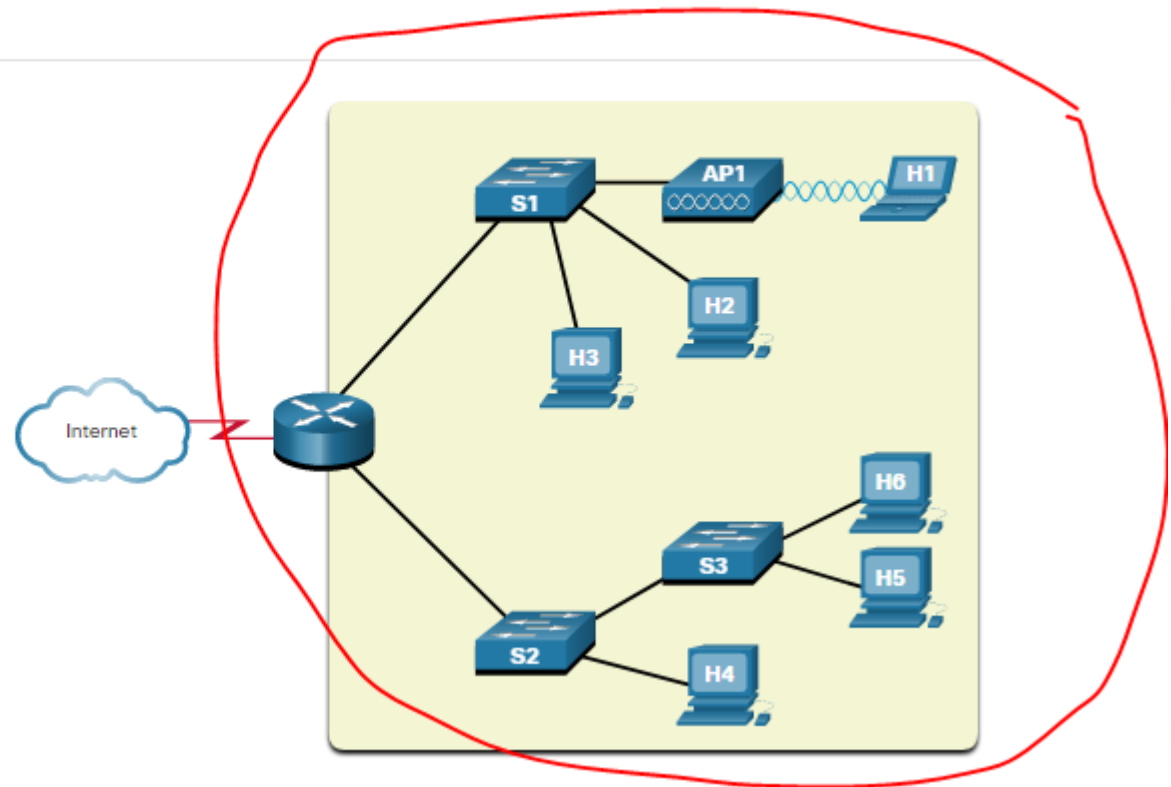
AP1

S1

S2

S3

Edge Router





Reduce Failure Domain Size

A well-designed network controls traffic and limits the size of failure domains (i.e., the area of a network that is impacted when the network experiences problems).

- In the hierarchical design model, failure domains are terminated at the distribution layer.
- Every router functions as a gateway for a limited number of access layer users.

Routers, or multilayer switches, are usually deployed in pairs in a configuration referred to as a building, or departmental, switch block.

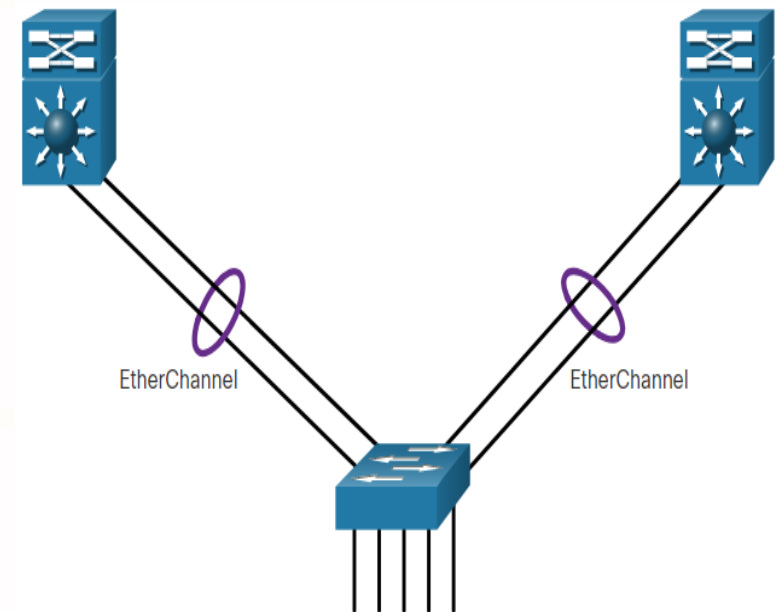
- Each switch block acts independently of the others.
- As a result, the failure of a single device does not cause the network to go down.



Increase Bandwidth

Link aggregation (e.g., EtherChannel) allows an administrator to increase the amount of bandwidth between devices by creating one logical link made up of several physical links.

- EtherChannel combines existing switch ports into one logical link using a Port Channel interface.
- Most configuration tasks are done on the Port Channel interface (instead of on each individual port) to ensure configuration consistency on the links.
- EtherChannel can load balance between links.





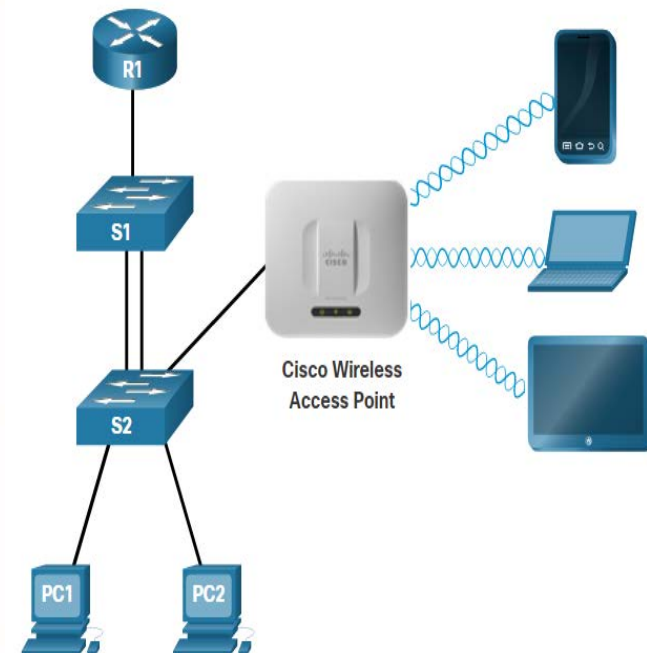
Expand the Access Layer

An increasingly popular option for extending access layer connectivity is through wireless.

- Wireless LANs (WLANs) provides increased flexibility, reduced costs, and the ability to grow and adapt to changing network and business requirements.
- To communicate wirelessly, end devices require a wireless NIC to connect to a wireless router or a wireless access point (AP).

Considerations when implementing a wireless network include:

- Types of wireless devices connecting to the WLAN
- Wireless coverage requirements
- Interference considerations
- Security considerations

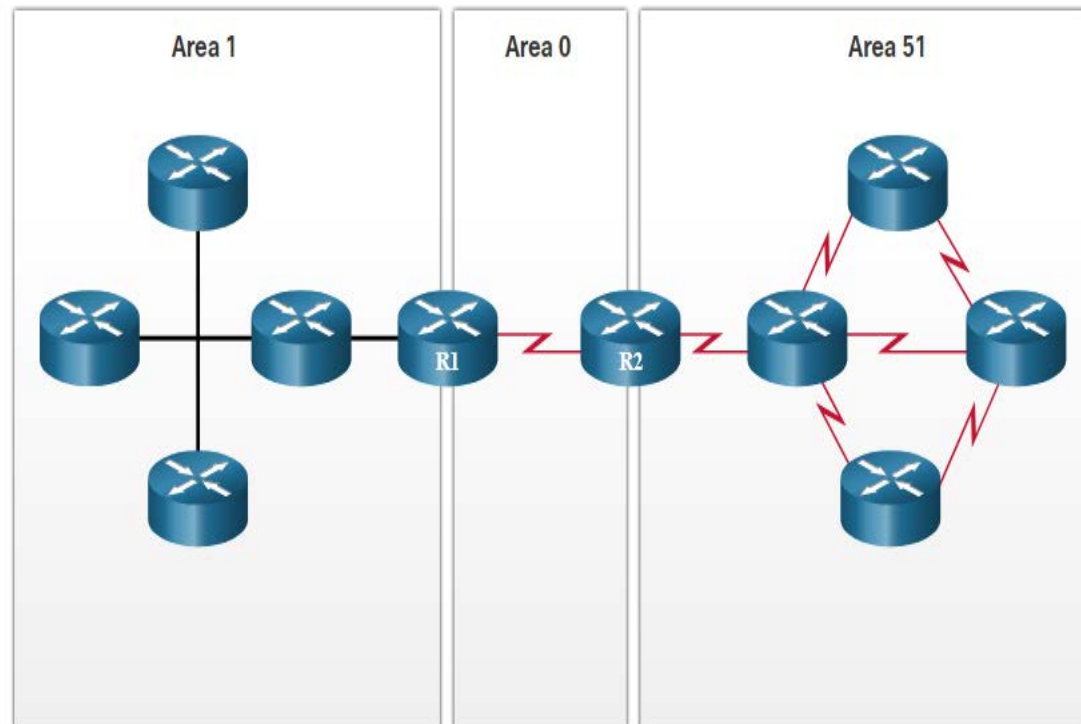




Tune Routing Protocols

Advanced routing protocols, such as Open Shortest Path First (OSPF) are used in large networks.

- OSPF is a link-state routing protocol that uses areas to support hierarchical networks.
- OSPF routers establish and maintain neighbor adjacencies with other connected OSPF routers.
- OSPF routers synchronize their link-state database.
- When a network change occurs, link-state updates are sent, informing other OSPF routers of the change and establishing a new best path, if one is available.





Multilayer Switching

Multilayer switches are typically deployed in the core and distribution layers of an organization's switched network.

- They support some routing protocols and forward IP packets at a rate close to that of Layer 2 forwarding.
- Multilayer switches often support specialized hardware, such as application-specific integrated circuits (ASICs).
- ASICs along with dedicated software data structures can streamline the forwarding of IP packets independent of the CPU.



VLANs

- Logical hierarchy imposed on a flat switched network allowing
 - Scalability
 - Formation of workgroups
 - Simplified admin
 - Better security



Summary

- Understand network traffic in terms of:
 - Flow
 - Load
 - Behavior
 - QoS requirements



Questions?